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REVIEW

AI-driven approaches for automatic detection of sleep apnea/hypopnea based on human physiological signals: a review

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Abstract

Sleep apnea/hypopnea is a sleep disorder characterized by repeated pauses in breathing which could induce a series of health problems such as cardiovascular disease (CVD) and even sudden death. Polysomnography (PSG) is the most common way to diagnose sleep apnea/hypopnea. Considering that PSG data acquisition is complex and the diagnosis of sleep apnea/hypopnea requires manual scoring, it is very time-consuming and highly professional. With the development of wearable devices and AI techniques, more and more works have been focused on building machine and deep learning models that use single or multi-modal physiological signals to achieve automated detection of sleep apnea/hypopnea. This paper provides a comprehensive review of automatic sleep apnea/hypopnea detection methods based on AI-based techniques in recent years. We summarize the general process used by existing works with a flow chart, which mainly includes data acquisition, raw signal pre-processing, model construction, event classification, and evaluation, since few papers consider these. Additionally, the commonly used public database and pre-processing methods are also reviewed in this paper. After that, we separately summarize the existing methods related to different modal physiological signals including nasal airflow, pulse oxygen saturation (SpO_2), electrocardiogram (ECG), electroencephalogram (EEG) and snoring sound. Furthermore, specific signal pre-processing methods based on the characteristics of different physiological signals are also covered. Finally, challenges need to be addressed, such as limited data availability, imbalanced data problem, multi-center study necessity etc., and future research directions related to AI are discussed.

Keywords: Sleep apnea/hypopnea, Automatic detection, AI-based techniques, Human physiological signals

Introduction

Sleep is an essential aspect of human life, providing the body with a much-needed opportunity to rest, recuperate, and recharge. The increasing focus on sleep has led to a growing interest in sleep staging [1] and the detection of sleep disorders in recent years. Sleep apnea and hypopnea are two prevalent sleep disorders that affect sleep patterns and respiratory function. Among them, sleep apnea characterized by repeated pauses in breathing during sleep. These pauses, known as apneas, can last for several seconds to even minutes and may occur multiple times throughout the night. This interruption in breathing is often caused by a partial or complete blockage of the airway, leading to a decrease in oxygen levels in the blood and a subsequent arousal from sleep to restore normal breathing. Sleep apnea can significantly impact the quality of sleep, leaving affected individuals feeling fatigued, irritable, and prone to daytime sleepiness. The classification of sleep apnea and rules for scoring respiratory are as follows:

(1) **Obstructive sleep apnea (OSA)**: OSA is a common disorder characterized by the pharyngeal muscles failing to keep the airway open during sleep [2]. The disease has already affected almost 1 billion people, with its prevalence exceeding 50% in some countries [3]. According to the American Academy

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of Sleep Medicine (AASM) [4], obstructive sleep apnea was defined as a ≥ 90% drop in respiratory flow for at least 10 s with breathing effort.

(2) **Central Sleep Apnea (CSA)**: CSA is uncommon and occurs due to a temporary failure in the pontomedullary pacemaker generating breathing rhythm [5]. CSA can result from certain medical conditions affecting the brainstem or medical treatments like opioid usage. The symptoms and health consequences of CSA are similar to those of OSA, including daytime sleepiness and increased cardiovascular risk. Central sleep apnea was defined as a ≥ 90% drop in respiratory flow for at least 10 s without breathing effort [6].

(3) **Mixed Sleep Apnea (MSA)**: MSA is a combination of both obstructive and central sleep apnea [4, 7]. This condition involves a combination of airway obstruction and insufficient respiratory effort from the brain. Mixed apnea was defined as a ≥ 90% drop in respiratory flow for at least 10 s with partial breathing effort.

As for hypopnea, it refers to shallow or excessively slow breathing during sleep, resulting in a reduction in airflow to the lungs. While it is not as severe as apnea, hypopnea can still disrupt sleep patterns and affect overall sleep quality. Like sleep apnea, it can lead to daytime sleepiness and other related health issues if left untreated. According to AASM [4], hypopnea is defined as follows: hypopnea is defined as a ≥ 30% drop in respiratory flow for at least 10 s, associated with ≥ 3% oxygen desaturation or arousal.

Both sleep apnea and hypopnea are commonly associated with snoring, a loud and sometimes disruptive sound caused by the vibration of the airway tissues due to restricted airflow. The prevalence of these sleep disorders is not limited by age, gender, or body size. Individuals with risk factors such as obesity, family history, and advanced age are particularly susceptible to an increased risk. Recognizing the signs and symptoms of sleep apnea and hypopnea is crucial for early diagnosis and effective treatment. Timely intervention can help improve sleep quality, enhance overall well-being, and reduce the risk of associated complications. The main manual scoring method is the apnea–hypopnea index (AHI). AHI provides a standardized way to quantify the number of apneas and hypopneas per hour of sleep. The severity of sleep apnea and hypopnea is determined by the AHI scores. The scoring rules are as follows: normal (AHI 5 events per hour(e/h)), mild (5≤AHI15 e/h), moderate (15≤AHI30 e/h) or severe sleep apnea (AHI≥30 e/h). The severity classification helps doctors determine the appropriate treatment approach for patients.

PSG, serving as the gold standard for diagnosing sleep apnea/hypopnea, plays a pivotal role in guiding effective management strategies to enhance both sleep quality and overall health among individuals with sleep-disordered breathing conditions [8–10]. As shown in Fig. 1, PSG data are usually collected in a sleep laboratory under the supervision of a qualified sleep technician. During sleep, a PSG device is used to gather data on multiple physiological parameters [11], enabling doctors to analyze various aspects of sleep architecture and identify abnormalities. And in the process, various signals can be recorded, including EEG, electrooculogram (EOG), electromyogram (EMG), ECG, oronasal airflow, SpO_2, respiratory movement and body position [12].

Traditionally, the diagnosis of sleep apnea and hypopnea has relied on expert experience and the collection of PSG data, often requiring patients to undergo testing in a hospital setting. However, this approach poses challenges as it necessitates the use of multiple sensors that significantly impact sleep quality and may yield inaccurate measurements, thereby affecting diagnostic accuracy. Moreover, manual scoring of sleep apnea and hypopnea events is time-consuming [13].

As shown in Fig. 2, the development of new technologies, such as novel sensors, Wi-Fi, remote monitoring, and wearable devices with photoplethysmography (PPG) sensors, has made it possible to utilize continuous pulse wave data during daily life [14]. This advancement enables home sleep apnea testing (HSAT), which can enhance accessibility for detecting sleep apnea/hypopnea and provide alternative options to polysomnography (PSG) [15]. Therefore, in recent years, many researchers have devoted their attention to developing automatic sleep apnea/hypopnea detection methods using single or multi-modal signals that could be provided by a portable home device.

The typical method for sleep apnea/hypopnea using the following procedures: (1) Pre-processing; (2) Feature

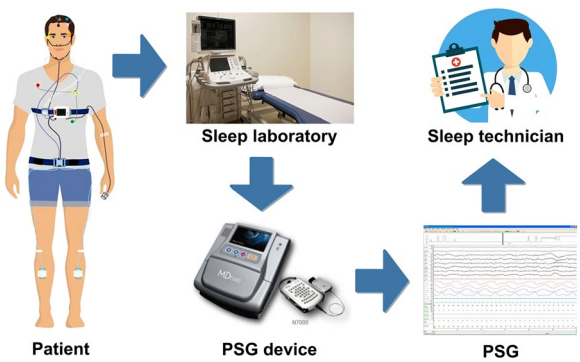


Fig. 1 PSG data acquisition for sleep apnea/hypopnea diagnosis

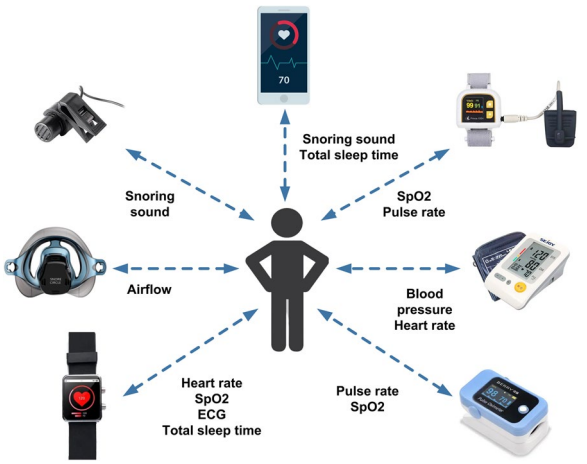


Fig. 2 Portable devices for diagnosing sleep apnea/hypopnea

extraction; (3) Feature selection; and (4) Classification. Particularly, the step of feature selection requires various domain expertise [16]. Deep learning models have the ability to analyze large amounts of complex data and automatically extract relevant features from raw data without redundant processes such as feature selection and domain knowledge [17]. This ability has led researchers to increasingly turn to building automated sleep apnea/hypopnea detection systems based on these models. For instance, Urtnasan et al. proposed a multiclass classification model to detect OSA based on six-layer Convolutional Neural Network (CNN). The dataset contains single-channel ECG signals collected from 87 patients [18]. Jiang et al. [19] designed a multi-scale parallel CNN to detect sleep apnea automatically by using single-channel EEG signals. For multi-modal approach, Memis et al. [20] proposed a model based on feature-level fusion of ECG and SpO_2 signals to effectively classify OSA.

Furthermore, some works implemented deep learning techniques on real-time sleep monitoring systems for sleep apnea/hypopnea detection in order to afford continuous assessment of individual's sleep patterns and respiratory for risk stratification in recent years. For example, Choi et al. [16] proposed a CNN-based approach for real-time apnea–hypopnea detection by using single-channel nasal airflow signal. The approach could be implemented as a supportive method to reduce event detection time in sleep laboratories.

In view of the fact that there is few work to summarize the automatic detection models of sleep apnea/hypopnea and their pre-processing methods for different modal signals, we summarize the typical flow diagram of an automatic diagnostic method for sleep apnea/hypopnea in Fig. 3. Most of works utilized the following 5 procedures: (1) Data acquisition; (2) Raw signal pre-processing; (3) Machine/deep learning model construction; (4) Classification; and (5) Evaluation. In the first step, data were mainly obtained from PSG devices in the laboratory or portable devices at home. The pre-processing step plays a crucial role in efficiently processing raw signals. Typically, the pre-processing step encompasses three key stages: denoising, normalization and segmentation. The image creation step is optional and is used in some studies because they believe that the one-dimensional (1-D) signal mainly captures temporal information but ignores frequency domain information. Considering the unique characteristics of each physiological signal modality, their respective pre-processing steps are tailored accordingly. For instance, heart rate variability (HRV) signals extracted from ECG signals and subband division of EEG signals, etc. This will be discussed in detail in the following sections. In addition, tasks including per-patient and per-segment classification, in the per-patient task, the severity classification including normal, mild, moderate and severe are calculated by AHI scores. Per-segment task is to classify the signal segments in order to test the performance of proposed methods. As for evaluation step, general metrics encompass sensitivity (true positive rate), precision (positive prediction value), specificity (true negative rate), accuracy (percentage of subjects correctly classified), F1-score (the harmonic mean of sensitivity and precision), and ROC curve.

Overall, the key contributions of this article are as follows:

- we do a comprehensive research on existing works from recent years about automatic diagnosis of sleep apnea/hypopnea approaches based on machine/deep learning methods that use single or multi-modal signals.
- We conclude the entire process of automatic sleep apnea/hypopnea detection, including data acquisition, raw signal pre-processing, machine/deep learning models construction, classification and evaluation.
- The challenges and solutions of AI-driven automated sleep apnea/hypopnea detection are thoroughly discussed, while future research directions are briefly summarized.

The structure of this paper is as follows: "Public database" section summarizes the most commonly used public databases; "Pre-processing methods" section states the general pre-processing step methods, which includes denoising, normalization, image creation; in "Feature extraction and model development methods" section, a

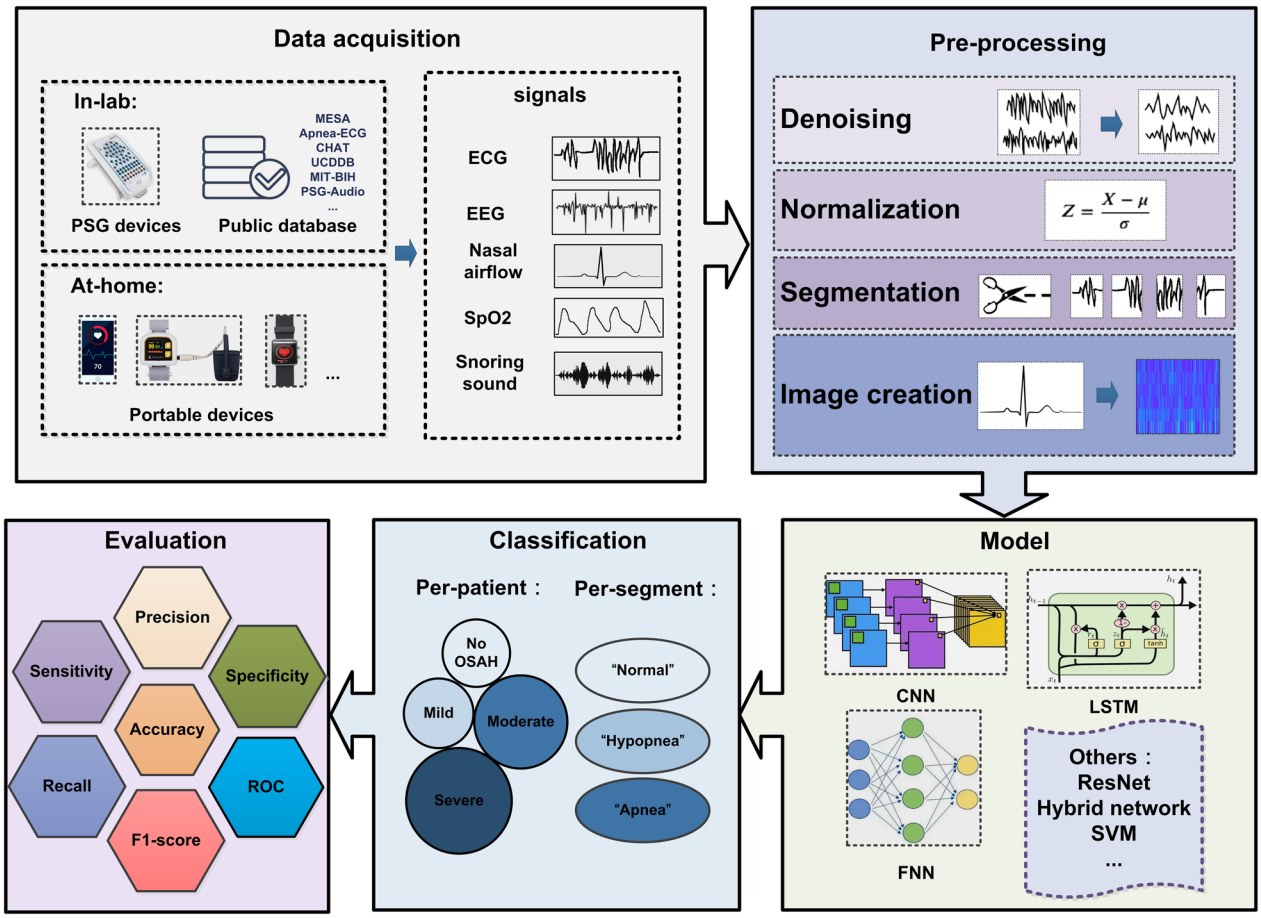


Fig. 3 The general architecture for automatic sleep apnea/hypopnea detection. The architecture consists of five parts: (1)data acquisition; (2)raw signal processing; (3)Model construction; (4) Classification; and (5) Evaluation

state of art-overview of literature related to the automatic sleep apnea/hypopnea detection based on machine/deep learning models that use nasal airflow, SpO_2, ECG, EEG and snoring sound signals are reviewed, and specific pre-processing methods for different modal signals are also mentioned; "Challenges and future work" section discusses the real-world challenges faced today and future work and "Conclusion" section concludes the paper.

Public database

In this section, we provide a comprehensive list of the most frequently utilized public databases in previous studies. These databases have primarily been sourced from hospitals or sleep laboratories and encompass participants spanning various ages, countries, and genders. Utilizing the same public database allows for performance comparisons among different methodologies.

(1) **MESA**: Multi-Ethnic Study of Atherosclerosisi (MESA) collected by the National Sleep Research Rsources (NSRR) [21, 22]. The database contains 2230 subjects aged from 54–93 years old. The PSG recordings had durations of at least 8 h and each recording is annotated by the sleep technologists as obstructive apnea, mixed apnea, and hypopnea with the starting time and duration of time. Overall, 15.0% of individuals had severe sleep disordered breathing (AHI ≥30); 30.9% had short sleep duration (6 h); 6.5% had poor sleep quality (sleep efficiency 85%); and 13.9% had daytime sleepiness. The dataset is available at: https://sleepdata.org/datasets/mesa

(2) **PhysioNet Apnea-ECG**: The database was provided by Dr. Thomas Penzel at Philipps University [23]. It contains 70 overnight ECG recordings collected from 25 male and 7 female subjects aged from 27–63 years old. The recordings are in two groups (a released set and a withheld set), each with 35 subjects. The ECG signals were recorded at a sampling rate of 100 Hz and with 16-bit resolution and

ranges from 6–8 h. Each recording contains a continuous digitized ECG signal, a collection of apnea and normal annotations (derived by sleep specialists from concurrently recorded respiration and associated data), and a set of machine-generated QRS annotations, in which all beats regardless of type have been labeled normal. In addition, eight recordings (a01-a04, b01, and c01-c03) are supplemented by four additional signals (Resp C and Resp A, chest and abdominal respiratory effort signals obtained using inductance plethysmography; Resp N, oronasal airflow measured using nasal thermistors; and SpO_2, oxygen saturation). The dataset is available at: https://physionet.org/content/apnea-ecg/1.0.0/

(3) **UCDDB**: The database [24] was provided by the Sleep Disorders Clinic at St. Vincent?s University Hospital, Dublin. It contains 25 complete overnight PSG recordings collected from 21 male and 4 female subjects aged from 28–68 years old with AHI scale ranges between 1.7 and 90.0. The recordings ranged from 6–8 h and were recorded at the sampling rate of 128 Hz. A collection of apnea events are manually annotated by sleep specialists according to standard scoring rules. The dataset is available at: http://bigd.big.ac.cn/databasecommons/database/id/5216

(4) **MIT-BIH**: The database was provided on CD-ROM [25], it contains 16 PSG recordings collected from 60 male subjects with or without sleep apnea syndrome aged from 32–56 years old. These recordings include EEG, EMG, EOG, invasive blood pressure, respiratory wave, SpO_2, and cardiac volume as measured by VEST method. The recordings ranged from 2–7 h, and were digitized at a sampling rate of 250 Hz and 12 bits. QRS waves were automatically detected and annotated by using screen editor. The dataset is available at: https://archive.physionet.org/physiobank/database/slpdb/

(5) **CHAT**: The childhood Adenotonsilectomy Trial (CHAT) database was provided by a Data Coordination Center (University of Pennsylvania; Philadelphia, PA) [26], which is a public multicenter database containing 1638 PSG recordings from 1232 subjects recruited between 2007 and 2012 aged from 5.0—10.0 years old with mild to moderate OSA. The dataset is available at: https://www.sleepdata.org/datasets/chat

(6) **PSG-Audio**: The database contains full-night PSG signals from 212 patients, containing all annotations by the medical team of the Sismanoglio-Amalia Fleming General Hospital of Athens [27, 28]. The snoring sounds were recorded together with the full-night PSG at the sleep disorders laboratory using a microphone placed over the trachel of the patients with the sampling rate of 48 kHz. The dataset is available at: https://www.scidb.cn/en/detail?dataSetId=778740145531650048

(7) **SHHS**: Sleep Heart Health Study (SHHS) database [29] is a prospective cohort study of the cardiovascular and other consequences of SDB. The database contains a variety of data on sleep health, including data related to arousal, sleep-disordered breathing, and more. All subjects were 40 years old or older with no history of sleep apnea treatment or ongoing home oxygen therapy at the baseline visit. The dataset was collected over a decade and contains two polysomnogram visits: SHHS1 (1995–1998), and SHHS2 (2001–2003), producing 5763 and 2651 patient records, respectively. Both the SHHS datasets include overnight SpO_2 and pulse rate (PR) signals with the sampling rate of 1 Hz. The dataset is available at: http://www.jhucct.com/shhs

Pre-processing methods

The methods employed for each pre-processing step, namely normalization, denoising, and image creation, are summarized in this section. Among them, the majority of the literature implements denoising and normalization steps. As for image creation, it is considered optional, mainly utilized by works employing two-dimensional (2-D) input models.

Denoising

Denoising is a crucial step of signal pre-prossing, the aim is to remove unwanted noise and disturbances from a signal and preserve the underlying useful information in order to enhance signal quality, facilitate machine learning, and improve analysis accuracy. Bandpass filter with efficient denoise ability that utilized by many works. It is actually an electronic circuit or a signal processing technique used to allow a specific range of frequencies to pass through while attenuating or blocking frequencies outside that range. It functions by letting signals within a certain frequency range, known as the "passband," passing through the filter while reducing the amplitudes of frequencies below the lower cutoff frequency and above the upper cutoff frequency, which together define the boundaries of the passband.

Normalization

Normalization is a critical data pre-processing technique utilized to standardize and transform features within a dataset, ensuring consistent scaling and distribution. This procedure offers numerous advantages, including

improved model performance and optimized gradient computation. It has been observed that the majority of previous studies employ Z-score normalization and Min-Max normalization as the preferred method for achieving this objective.

Z-score normalization

Z-score normalization is a statistical approach employed to convert data into a standardized distribution with a mean of 0 and a standard deviation of 1 [30]. The formula for calculating the Z-score of a data point is as follows:

$$Z = \frac{X - \mu}{\sigma} \tag{1}$$

where, Z is the Z-score of the data point, X is the original value of the data point, μ is the mean of the dataset, and σ is the standard deviation of the dataset. The resulting Z-score indicates how many standard deviations the data point is away from the mean. A positive Z-score indicates that the data point is above the mean, while a negative Z-score indicates that it is below the mean.

Min–Max normalization

The linear function transforms the original data linearization method to the range [0,1], and the normalization formula is as follows:

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}} \tag{2}$$

Where, X is the original dataset, X_{max} and X_{min} are the maximum and minimum values of the original dataset, respectively.

Image creation

Since most of the raw physiological signals are 1-D sequences, which only reflect time-domain information and ignore frequency-domain information. Many researchers tend to convert 1-D signal into image with both time-frequency information for their models to effectively process and analyze. The image creation methods commonly used in previous work are listed below.

Continuous wavelet transform (CWT)

Continuous wavelet transform provides a time-frequency representation (often referred to as a scalogram) of the signal, showing how its frequency content changes overtime. The main idea of CWT is to analyze a signal by comparing it to a family of wavelet functions, each of which is a scaled and translated version of a basic wavelet function. These wavelets are typically short in time duration and can capture the characteristics of different frequency components of the signal at different time scales [31]. Mathematically, the CWT of a signal $f(t)$ with respect to a wavelet $\psi(t)$ is given by:

$$CWT_f(a,b) = \int_{-\infty}^{\infty} f(t) \frac{1}{\sqrt{a}} \psi^* \left(\frac{t-b}{a} \right) dt \tag{3}$$

where, a is the scaling parameter that controls the width of the wavelet. It is responsible for changing the time scale of the analysis. b is the translation parameter that determines the position of the wavelet along the signal. $\psi^*(t)$ is the complex conjugate of the wavelet function $\psi(t)$.

Short time Fourier transform (STFT)

Short Time Fourier Transform provides a 2D representation of physiological signal (often referred to as a spectrogram) with a constant time-frequency resolution, which is created by a constant sliding window along the time axis [32]. Mathematically, the STFT is defined as follows:

$$STFT(t,f) = \int_{-\infty}^{\infty} x(\tau) h(\tau - t) e^{-j2\pi f \tau} \, d\tau, \tag{4}$$

where, $h(\tau - t)$ is window function

Gabor transform (GT)

GT is a complex-valued function that is particularly well-suited for analyzing the time-frequency characteristics of signals and images. It utilizes a Gaussian window to clip out a piece of time-domain signal, before applying the Fourier transform [33]. It looks like symmetric, identical bandpass filters separated evenly in the frequency domain. The Time-Frequency spectral image of GT is called a spectrogram. The GT of a given input signal $s(t)$is evaluated as follows:

$$G_s(t,f) = \int_{-\infty}^{\infty} s(\tau) g_{t,f}^*(\tau) \, d\tau \tag{5}$$

where

$$g_{t,f}(\tau) = g(\tau - t) e^{j2\pi f.t} \tag{6}$$

and $*$ indicate the conjugate transpose. Since the window function $g(t)$ of GT is a Gaussian window function $e^{-\pi t^2}$, thus $G_s(t,f)$ is expressed as:

$$G_s(t,f) = \int_{-\infty}^{\infty} s(\tau) e^{-\pi(\tau - t)^2} e^{-j2\pi f.t} \, d\tau \tag{7}$$

The hypothesis is that $s(t)$ should be almost steady across the time window $g_{t,f}\tau$, and the window must be of the same length [34].

Feature extraction and model development methods

In this section, we summarize the automatic diagnosis methods in existing literatures by using different modal physiological signals, including nasal airflow, SpO_2, ECG, EEG, and snoring sound. Tables 2, 3, 4, 5 and 6 briefly summarize the existing techniques reviewed in this paper according to signal category, respiratory events, methods, performance, database, subjects, and year. Additionally, in these tables, "N" represents normal event, "A" represents sleep apnea (including obstructive, central, and mixed sleep apnea), and "H" represents hypopnea. The "/" represents the classification categories. Table 1 shows the nomenclature of sleep apnea/hypopnea datasets used in reviewed literatures in this paper.

Airflow signal

Figure 4 shows the waveform of three classes of nasal pressure airflow including normal, hypopnea, and apnea. Respiration signals are the most related signals to sleep syndrome. They are used both in clinical studies and in engineering for sleep syndrome analysis [35]. And the nasal airflow signal can be recorded using a variety of methods, including nasal cannulae or pressure sensors placed near the nostrils. According to AASM [4], the nasal airflow is the recommended signal for the detection of apnea/hypopnea. The scoring rules for sleep apnea:

(1) There is a drop in the peak thermal sensor excursion by $\geq$ 90% of baseline.
(2) The duration of the event lasts at least 10 s.
(3) At least 90% of the event's duration meets the amplitude reduction criteria for apnea.

The scoring rules for sleep hypopnea:

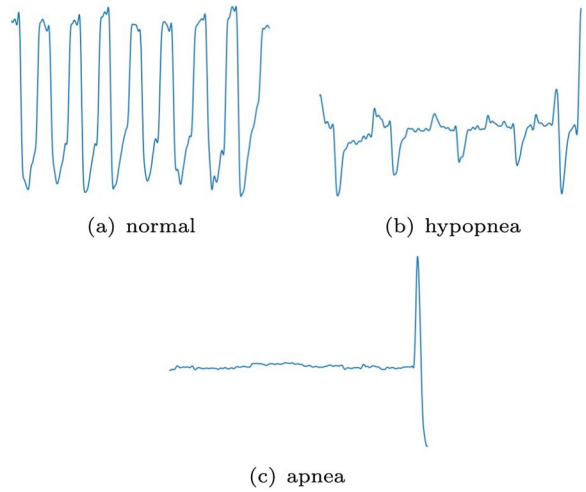


Fig. 4 Three categories of nasal airflow waveform: **a** normal waveform, **b** hypopnea waveform and **c** apnea wave form

(1) The peak signal excursions drop by $\geq$ 30% of pre-event baseline using nasal pressure.
(2) The duration of the $\geq$ 30% drop in signal excursions is $\geq$ 10 s.
(3) There is $\geq$ 3% oxygen desaturation from the pre-event baseline or the event is associated with an arousal.

Deep learning techniques for automated sleep apnea and hypopnea detection using nasal airflow are summarized in Table 2. It is worth noting that some studies have focused on transforming the 1-D raw nasal airflow signal into a 2-D input, such as a time-frequency image, before constructing the processing model. Consequently, this article presents separate descriptions for methods utilizing both 1-D and 2-D inputs.

Table 1 Nomenclature of sleep apnea/hypopnea database

MESA	Multi-Ethnic Study of Atherosclerosisi
Apnea-ECG	PhysioNet Apnea-ECG Dataset
UCDDB	St. Vincent's University Hospital/University College Dublin sleep apnea database
CHAT	The Childhood Adenotonsilectomy Trial
MIT-BIH	MIT-BIH Polysomnographic Dataset
PSG-Audio	Sismanoglio-Amalia Fleming General Hospital of Athens Dataset
FAH	The Sleep Center of the First Affiliated Hospital, Sun Yat-sen University.
CMH	The Integrative Department of GuangDong Province Traditional Chinese Medical Hospital
UofC	The University of Chicago Medicine
BUH	Burgos University Hospital
SHHS	Sleep Heart Health Study
SUMS	The Shiga University of Medical Science Hospital

Table 2 Methods related to nasal airflow

Study	Signals	Events	Methods	Performance (%)	Subjects	Database	Year
[36]	Nasal airflow	N/A/H	BiLSTM	Accuracy = 98.18	100	MESA	2022
[37]	Oronasal thermal airflow, nasal pressure, abdominal respiratory inductance plethysmography	N/A	LSTM/BiLSTM	LSTM: Accuracy = 85.4, BiLSTM: Accuracy = 85.6	17	Interdisciplinary Center of Sleep Medicine in Charité- Universitätsmedizin Berlin.	2020
[35]	Nasal airflow, nasal temperature	N/A/H	LSTM	Clinical: Accuracy = 81.6	144	MIH-BIH, Clinical database	2019
[38]	Nasal airflow	N/A/H	2-D CNN	Accuracy = 79.8	1507	MESA	2018
[39]	Nasal airflow	N/A/H	Multi-resoluation ResNet	Accuracy = 91.2	450	FAH, CMH	2021
[40]	Nasal airflow	N/A/H	ResNet	Accuracy = 91.23	500	FAH, CMH	2021
[41]	Nasal airflow, SpO_2	AHI severity	2-D CNN	CHAT: Accuracy = 72.55, UofC: Accuracy = 61.79	2612	CHAT, UofC	2022
[42]	Nasal airflow, abdominal, thoracic	n/A/H	CNN, Markov chain	Accuracy = 80.78	1507	MESA	2020
[43]	Nasal airflow, SpO_2	N/A/H	Feature-fusion 1-D CNN	Accuracy = 95.91	450	FAH, CMH	2024

Table 3 Methods related to SpO_2

Study	Signals	Events	Methods	Performance(%)	Subjects	Database	Year
[45]	SpO_2	AHI Severity	1-D CNN	Accuracy = 88.3	2047	HSAT	2021
[46]	SpO_2	AHI Severity	1-D CNN	CHAT: Accuracy = 72.8, UofC: Accuracy = 60.2, BUH: Accuracy = 60	3196	CHAT, UofC, BUH	2021
[47]	SpO_2	N/A	1-D CNN	Accuracy = 97.08	25	UCDDB	2021
[48]	SpO_2	N/A	SVR, DT, MLP	Accuracy = 91	1305	Beijing Children's Hospital Sleep Center	2022
[49]	SpO_2, ECG	N/A	ANN	Accuracy = 91.83	8	Apnea-ECG	2022
[50]	SpO_2, ECG	N/A	FNN	Accuracy = 87.41	8	Apnea-ECG	2022
[51]	SpO_2, ECG	N/A/H	SVM	Accuracy = 99.1	13	Apnea-ECG, Hallelujah General Hospital Sleep Laboratory	2021
[52]	SpO_2, PR	NA/H	1-D CNN	SHHS1: Accuracy = 84.3, SHHS2: Accuracy = 82.2	8068	SHHS	2022

SVR support vector regression, *DT* decision tree

Methods for 1-D inputs

Using LSTM and BiLSTM The Long Short-Term Memory (LSTM) neural network framework is an architectural design that incorporates feedback loops to effectively process sequential data, addressing the issue of vanishing gradients and capturing long-term dependencies within sequences [44]. Given that airflow signals are 1-D data, LSTM-based models have been widely employed for their suitability in processing such data. In a recent study by Jansri et al. [36], they utilized Bidirectional Long Short-Term Memory (BiLSTM) to extract temporal features from single-channel signals for sleep apnea and hypopnea detection. Compared to LSTM, BiLSTM enhances classification performance by incorporating both past and future features through its backward direction with reverse time order. This model achieved a remarkable accuracy of 98.18% in detecting apnea and hypopnea. Additionally, nasal airflow can be accompanied by other PSG data for the detection of apnea/hypopnea. The utilization of multi-modal signals could provide multi-variate feature information to enhance the efficiency of classifying respiratory events. In [37], the authors employed oronasal thermal airflow signal (FlowTh), nasal pressure (NPRE), and abdominal respiratory inductance plethysmography (ABD) signals to feed a two-layer BiLSTM-based model, achieving an average accuracy of 85.6%. They also evaluated the LSTM model which had a slightly lower accuracy of 85.4% compared to BiLSTM. The study conducted by Yang et al. [35] utilized two respiration channels, namely nasal airflow and nasal temperature, as inputs for an LSTM-based model designed to automatically detect apnea and hypopnea from PSG recordings. The overall accuracy achieved on the two databases was 81.6%.

Using 1-D CNN Haidar et al. [42] introduced a novel approach for detecting obstructive apnea and hypopnea

using a Markov Chain augmented-CNN model, which incorporated three channels of respiratory signals (nasal airflow, abdominal, and thoracic). By combining the probabilities derived from the Markov Chain with CNN predictions, the model accurately classified the type of event in 30-second-long segments with an achieved accuracy of 80.78%. In addition, Peng et al. [43] considered combining nasal airflow with SpO_2 for automatic detection of sleep apnea/hypopnea. They proposed a bimodal modal feature fusion CNN model, which mainly consists of a dual-branch CNN module and a feature fusion module for classifying 10-second long nasal pressure airflow segments and SpO_2. Secondly, an efficient channel attention mechanism (ECA) is introduced in the second module, which adaptively weights the feature maps of each channel to improve classification accuracy. After 5-fold cross validation, the model achieved an excellent accuracy of 95.91 %.

Methods for 2-D inputs

Using 2-D CNN The 2-D Convolutional Neural Network (2-D CNN) is an advanced version of the conventional CNN architecture, specifically designed for processing and analyzing 2-D data, particularly images. In their study [38], the authors employed CWT with the analytical Morlet wavelet to obtain a 2-D image wavelet spectrogram. This spectrogram effectively visualizes the frequency spectrum and captures variations in events occurring at different frequencies by transforming 1-D nasal airflow signals. The authors proposed a 2-D CNN model to automatically detect apnea and hypopnea events by classifying spectrograms. Experimental results demonstrated that the accuracy of the 2-D CNN model was 79.8%, outperforming the 1-D CNN model (77.6% accuracy), which utilized raw nasal airflow signals as input. Additionally, it is worth noting that the versatility of the 2-D CNN extends beyond processing single-modal signals. In [41], the authors introduced a novel approach utilizing a 2-D CNN for processing both airflow and SpO_2 signals. These signals were segmented into 300-second-long segments and transformed into 2-D tensors with dimensions of size 3000*2, serving as inputs for the proposed network architecture. In this method, four classes representing AHI severity levels (normal, mild, moderate, or severe) were estimated. The achieved accuracy on CHAT dataset reached up to 72.55%, while on the clinical dataset it reached up to 61.79%. Overall, these findings highlight the effectiveness of employing a specialized deep learning framework like the proposed 2-D CNN for accurately detecting respiratory events and estimating AHI severity levels.

Using ResNet Residual Network (ResNet) is a deep convolutional neural network architecture that utilizes residual blocks to effectively mitigate the vanishing gradient problem and enable deeper network training, making it highly suitable for image classification tasks. Yue et al. [39] proposed a multi-resolution ResNet model that utilizes airflow spectrograms obtained through STFT to detect obstructive apnea and hypopnea. The model captures the multi-resolution imaging features of the airflow, achieving a high accuracy of 91.2% in AHI estimation. Similarly, the authors in [40] also utilized STFT and proposed a CNN-based model with residual blocks and a small group of filters with large kernel size, which also achieved a high accuracy of 91.23%.

SpO_2 signal

The application of pulse oximetry could estimate AHI to determine the severity of sleep apnea and hypopnea in an individual [53]. Furthermore, the inconvenience of recording full PSG data can be replaced by recording overnight continuous SpO_2 signal with a finger pulse oximetry measurement setup [54]. Three classes of SpO_2 signals are shown in Fig. 5. The machine/deep learning techniques employed for analyzing SpO_2 signals are summarized in Table 3. We can find that several studies have explored the combination of SpO_2 with other modal signals, which has proven advantageous in enhancing model accuracy. Consequently, methods for single-modal and multi-modal inputs are presented separately below.

Methods for single-modal inputs

Using 1-D CNN In [45], the authors used a 1-D CNN to screen sleep apnea for patients with acute cerebrovascular disease. 10-min-long 0.5 Hz SpO_2 segments were as the inputs of a 4-layer-CNN. For HSAT dataset without EEG signal, the respiratory event index (REI) which represents the number of apnea and hypopnea events per hour was estimated in this method, the results of

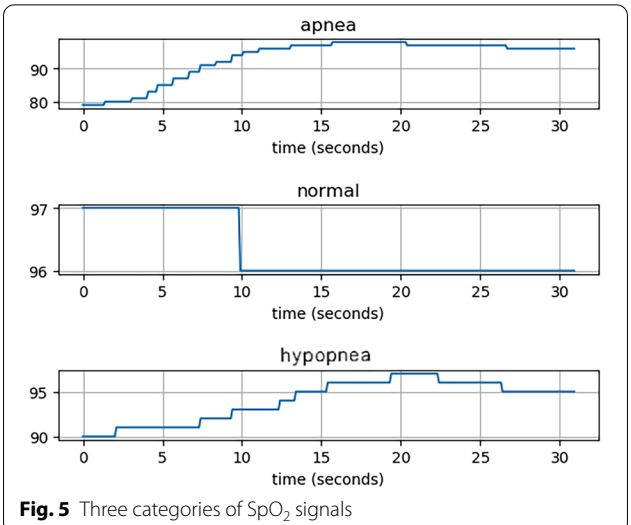


Fig. 5 Three categories of SpO_2 signals

four classes apnea severity (normal, mild, moderate, and severe) reached the accuracy of 88.3%. Vaquerizo et al. [46] also developed a 1-D CNN to estimate the severity of four categories of AHI, using three independent datasets: CHAT with 1638 subjects, UofC with 980 subjects, and BUH with 578 subjects. To ensure frequency homogeneity, the SpO_2 signals were down-sampled to 1 Hz and segmented into 20-min-long segments for processing. The CNN model achieved an accuracy of 72.8% in CHAT, 60.2% in UofC, and 60% in BUH. John et al. [47] proposed a novel approach named SomnNET, which utilizes a 1-D CNN to detect sleep apnea on a per-second basis using 11-s-long SpO_2 signals sampled at 8 Hz. This method achieved an impressive accuracy of 97.08%, surpassing several previous lower resolution apnea detection methods.

Using MLP Wu et al. [48] proposed a Multilayer Perceptron (MLP) neural model consisting of three distinct neuron layers (input, hidden, and output) for the detection of sleep apnea using single-channel SpO_2 signals from 1305 subjects. This model demonstrated high accuracy in diagnosing moderate to severe OSA but exhibited limited efficacy in diagnosing mild OSA.

Methods for multi-modal inputs

Moreover, SpO_2 signal is often accompanied by other signals such as ECG, nasal airflow, and PR to improve inferences.

Using 1-D CNN The study conducted by Sharma et al. [52] utilized both SpO_2 and PR signals for the detection of sleep apnea. During the data pre-processing stage, PR samples were collected within a range of 40–150 beats per minute. A 1D CNN model with Lenet-like architecture was designed and achieved impressive test performance, with an area under the receiver operating characteristic curve (ROCAUC) of 90.4%.

Using ANN and FNN Paul et al. [49] investigated a real-time sleep apnea detection system using SpO_2 and ECG signals individually, as well as their combination. The inputs of the FeedForward Artificial Neural Network (ANN) based model for sleep apnea detection were 1-min-long SpO_2 and R–R interval signals (the time interval between two successive R-peaks). Each model consisted of an input layer, five hidden layers with 100, 50, 25, 10, and 5 neurons respectively, and an output layer. Model performance evaluation was conducted using a 10-fold cross-validation approach. Test results demonstrated that the combined signal-based model achieved an accuracy of 91.83%, outperforming the individual signal-based models with accuracies of 80.04% for ECG only and 90.78% for SpO_2 only. Similarly, in [50], a three-layer Feedforward Neural Network (FNN) model was proposed to detect sleep apnea based on SpO_2 signals and R–R interval series. This model using multi-sensor signals obtained a higher accuracy of 87.41% than single-modal signal models which achieved accuracies of 61.28% using ECG only and 62.78% using SpO_2 only.

Using SVM In [51], this paper proposed an automatic sleep apnea–hypopnea detection system based on support vector machine (SVM). Various features from R–R intervals of ECG and a statistical feature from SpO_2 were extracted for training SVM. This model showed good performance with an accuracy of 99.1%. The model achieved better accuracy by utilizing a combination of bi-sensor signals compared to single-modal signal.

ECG signals

The physiological state of the heart over a period is recorded by an ECG [73]. The ECG signal has been widely utilized in various studies due to its ability to reflect respiratory rhythm. In fact, deep learning models can extract feature maps that capture both visible and invisible fluctuations in single-channel ECG signals caused by sleep apnea and hypopnea [18]. Figure 6 illustrates the waveforms of three categories of ECG signals (normal, hypopnea, and apnea). It can be observed that compared to the apnea and hypopnea classes, the R amplitudes of the normal class exhibit more significant fluctuations, which accurately reflect respiratory rhythm.

The pre-processing steps for the raw ECG data are diverse, as depicted in Fig. 7. Sleep apnea–hypopnea can cause heart rate variations [74], and sleep-related HRV, including R–R intervals and amplitudes, is a common characteristic signal for detecting sleep apnea/hypopnea. The combination of ECG and sleep-related HRV has the potential to predict OSA [75]. In previous studies, 11 related features such as meanNN, SDNN, Total Power,

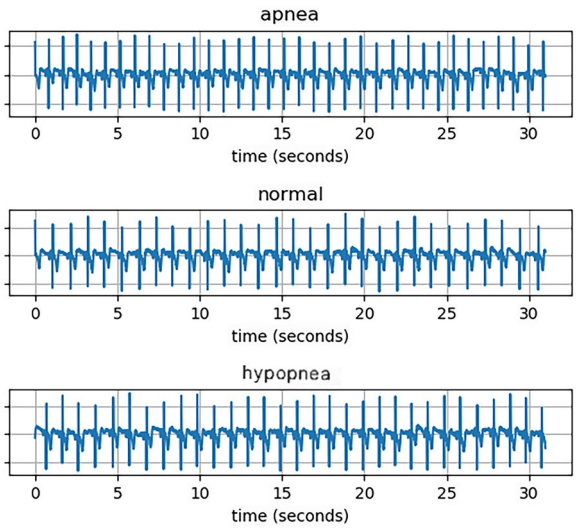


Fig. 6 Three categories of ECG signals

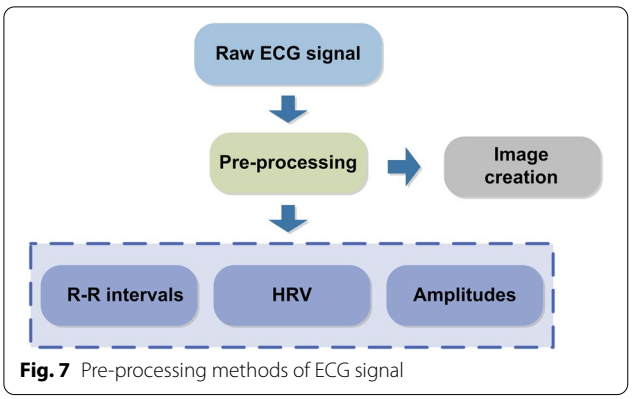


Fig. 7 Pre-processing methods of ECG signal

RMSSD, NN50, pNN50, LF, HF, LF/HF, LFnu, and HFnu have been commonly utilized [76, 77].

Table 4 summarizes the deep learning techniques used for ECG. Considering that a significant number of studies tend to convert the 1-D raw ECG signal into a 2-D input, such as a time-frequency image, before constructing the processing model. Therefore, we have categorized and presented methods utilizing both 1-D and 2-D inputs separately below.

Methods for 2-D inputs

Due to the fact that time-frequency images are commonly employed for signal learning purposes, several studies convert raw ECG signal into spectrogram using CWT, STFT, or other image creation techniques.

Using 2-D CNN As one of the most suitable methods for processing image data, 2-D CNN has been used in many works [55–57]. In [55], the authors developed a novel scalogram-based convolutional neural network (SCNN) to detect OSA using single-channel ECG signal. CWT was used to convert ECG signal into scalogram. In addition, empirical mode decomposition (EMD) was applied to the signals to find corrected intrinsic mode functions (IMFs), and then apply CWT on the IMFS to get hybrid scalograms. A lightweight CNN model was trained on ECG-only or hybrid scalograms for OSA detection. This model showed great performance on Apnea-ECG dataset with the accuracy of 94.30% in per-segment classification. Moreover, In [57], a deep learning model using smoothed Gabor spectrogram (SGS) of an ECG signal called OSACN-Net is proposed for automated OSA detection to obtain high performance. The proposed model finally achieved an average classification accuracy of 94.81% using SGS under a 10-fold cross-validation strategy. Compared to Squeeze-Net and ResNet-50, the developed OSACN-Net is more accurate and lightweight as it requires few learnable parameters,

Table 4 Methods related to ECG

Study	Signals	Events	Methods	Performance(%)	Subjects	Database	Year
[55]	ECG	N/A	2-D CNN	Apnea-ECG: Accuracy = 94.38, UCDDB: Accuracy = 81.86	95	Apnea-ECG, UCDDB	2021
[56]	ECG	N/A	2-D CNN	Accuracy = 92.41	70	Apnea-ECG	2021
[57]	ECG	N/A	2-D CNN	Accuracy = 94.81	70	Apnea-ECG	2022
[58]	ECG	N/A	2-D CNN	Accuracy = 93.33	70	Apnea-ECG	2023
[59]	ECG	N/A	1-D CNN	Accuracy = 88.23	70	Apnea-ECG	2020
[60]	ECG	N/A	SE-MSCNN	Accuracy = 90.64	70	Apnea-ECG	2022
[61]	ECG	N/A	1-D CNN	Accuracy = 89.4	70	Apnea-ECG	2021
[62]	ECG	N/A	LSTM	Sensitivity = 91	58	SUMS	2021
[63]	ECG	N/A	LSTM	Accuracy = 99.8	35	Apnea-ECG	2021
[64]	ECG	N/A	CNN-LSTM	Apnea-ECG: Accuracy = 97.21, UCDDB: Accuracy = 93.7	60	Apnea-ECG, UCDDB	2022
[65]	ECG	N/A	CNN-LSTM	Accuracy = 90.92	70	Apnea-ECG	2021
[66]	ECG	N/A	CNN-LSTM	Accuracy = 94.95	70	Apnea-ECG	2022
[67]	ECG	N/A	ResNet	UCDDB: Accuracy = 75.1, Apnea-ECG: Accuracy = 90.3	95	Apnea-ECG, UCDDB	2022
[68]	ECG	N/A	ResNet	Accuracy = 93.05	70	Apnea-ECG	2021
[69]	ECG	N/A	CNN-Transformer	Accuracy = 88.2	70	Apnea-ECG	2023
[70]	ECG	N/A	Hybrid Transformer	Accuracy = 91	70	Apnea-ECG	2022
[71]	ECG	N/A	AE	Accuracy = 85.1	70	Apnea-ECG	2021
[72]	ECG, SpO_2, BMI	N/A	FNN	Accuracy = 97.8	148	Beijing Tongren Hospital	2021

BMI body mas index

which makes it computationally fast and efficient. Niroshana ret al. [56] proposed a novel fused-image-based technique that detects OSA using single-channel ECG signal. A CNN model extracts features automatically from images created with 1-min ECG segments. The blending scalogram and spectrogram images further improved the system's detection abilities. The results of this study demonstrated that the proposed classifier can perform OSA detection with an average accuracy of 92.4% for the fused spectral images. In [58], this paper adapted a pre-trained 2-D CNN for OSA classification. The ECG signals were split into 2-second-long segments and subsequently filtered. Then, the recurrence plots (RP) algorithm was employed to transform these segments into spectrogram images. Notably, the inception-v3 model achieved an outstanding accuracy rate of 93.33%.

Methods for 1-D inputs

Using 1-D CNN In [59], HRV data (R–R intervals) were computed in the raw ECG signal and fed into the 1-D CNN based model for automatic sleep apnea detection. R–R intervals are the features because HRV is an evently sampled data and the number of R–R intervals in each 5-min-long segment. The proposed method achieved an accuracy of 88.23%. Further more, in [60], a lightweight sleep apnea detection method based on a multi-scaled fusion network named SE-MSCNN using single-channel ECG signal was developed. The proposed SE-MSCNN mainly includes a multi-scaled CNN module and a channel-wise attention module. In order to facilitate sleep apnea detection performance, various scaled ECG information with different length adjacent segments was extracted by three subneural networks. To overcome the problem of local concentration of feature fusion with concatenation, a channel-wise attention module with a squeeze-to-excitation block was employed to fuse the different scaled features adaptively.

Using LSTM The authors in [62] also used the HRV signal extracted from single-channel ECG recordings, the authors proposed a sleep apnea syndrome (SAS) screen method based on LSTM. Of course, for raw 1-D ECG signals, LSTM is the suitable model for processing time series signals. In [63], this paper used LSTM based model to detect sleep apnea in R–R intervals. The R–R intervals were generated by automatic QRS detection from Apnea-ECG benchmark database [78]. 10-fold cross-validation approach indicated that the LSTM network had 99.80% accuracy.

Using CNN-LSTM Recently, many works have constructed hybrid network called CNN-LSTM based on a combination of CNN and LSTM architecture. The CNN-LSTM hybrid network combines the advantages of CNN and LSTM and is useful for tasks involving spatial and temporal dependencies in data.

In [65], the authors proposed a novel approach based on CNN-LSTM for automatic sleep apnea detection. The hybrid network consists of two blocks of 1-D CNN and a block of two-layer LSTM. Among them, useful features from R–R interval and R amplitude can be automatically extracted by this hybrid model without feature engineering. For the testing dataset, the accuracy achieved by the proposed model was 90.92%. Similarly, reference [64] developed an automatic OSA detection method by building CNN-LSTM. The result accuracy was 97.21% in Apnea-ECG database and 93.70% in UCDDB. Moreover, in [66], ECG R-peak amplitudes and R–R intervals were extracted and aligned using power spectral analysis, and a CNN-LSTM model was developed to extract the most predictive ECG features and predict the occurrence of apnea. The prediction accuracy was as high as 94.95%, which was higher than the performance of conventional multilayer perceptrons (p0.05).

Using ResNet Although there have been several works implementing CNN-based model to process raw 1-D ECG signal, there is a problem that the convergence of the model starts to degrade with a deep architecture. Therefore, using residual learning such as identity and projection shortcuts is a suitable approach [79]. In reference [68], a 1-D CNN with residual connections was proposed for automatic apnea detection by using single-channel ECG signals. In addition, bayesian optimization for fine-tuning the network hyperparameters was also employed. The proposed method achieved an accuracy of 93.05% in detecting sleep apnea segments. Furthermore, Yang et al. [67] proposed a 1-D squeeze-and-excitation (SE) residual group network to thoroughly extract the complementary information between HRV and ECG-derived respiration (EDR), considering that most methods focused on HRV but ignored EDR. The accuracy of per-segment detection in Apnea-ECG was 90.3%.

Using Transformer Transformer structure with a self-attention mechanism [80] is suitable to be implemented in signal processing, where there is a meaningful dependency between different sequences. In [69], this paper proposed a CNN-Transformer architecture for automatic OSA detection based on single-channel ECG signal. The proposed architecture has two fundamental parts: the first part has the aim of learning a feature representation from ECG signal by using the CNN; The second part consists mainly of the transformer, a model structure built solely with self-attention mechanism, which is used to model the global temporal context and to perform classification tasks. The per-segment classification accuracy reached 88.2%, and the area under the receiver operating characteristic curve (AUC) was 0.95. The per-recording

classification accuracy reached 100% and the mean absolute error (MAE) was 4.33.

In [70], Hu et al. proposed a hybrid transformer model based on a self-attention mechanism for OSA detection by using single-channel ECG signal. Four types of inputs, including raw ECG signal, R-peak amplitude (RA), interbeat interval (RRI) sequence and RR interval first-order difference (RRID) sequence. The whole architecture consists three main parts:(1) A multi-perspective channel attention (MPCA) block, which is proposed to extract multi-perspective features; (2) Transformer block, which is designed to process the output features of MPCA with position encodings, in order to encode the most important information with self-attention mechanism; (3) The linear layer, which outputs the prediction results. This model achieved a high accuracy of 91%.

Using AE An autoencoder (AE) is a type of neural network used for unsupervised learning and dimensionality reduction. It is designed to learn a compressed representation of the input data, which is then used to reconstruct the original data as accurately as possible [81]. In [71], this paper developed an unsupervised sleep apnea detection model. The model was constructed based on a frequential stacked sparse auto-encoder (FSSAE) and a time-dependent cost-sensitive (TDCS) classification model. Where, the FSSAE could extract feature sets automatically with unsupervised learning technique. And the TDCS classification model was composed of a hidden Markov model (HMM) and the MetaCost algorithm, which could improve the performance of the classifier by considering temporal dependence and the imbalance problems. The classification result of per-segment reached an accuracy of 85.1%.

EEG signals

The EEG signals serve as crucial tools in the investigation of brain events, functions, and disorders [92–95]. They are the spontaneous potential activities generated by brain neural activity and always present within the central nervous system [96]. Furthermore, EEG signals contain much important information, which reflects brain nerve activity and abnormal breathing. There are three classes (normal, apnea, hypopnea) of EEG signals in Fig. 8. In recent years, many researchers applied EEG signals to sleep apnea–hypopnea detection. As is shown in Fig. 9, subband signal divided is crucial in raw EEG signal pre-processing steps. EEG signals could be categorized into five frequency subbands, namely delta (0–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–32 Hz), and gamma (32–40 Hz) [97]. The occurrence of apnea–hypopnea may lead to associated changes in these subband waves [98, 99]. Based on the aforementioned characteristics, a set of features can be extracted from five subband

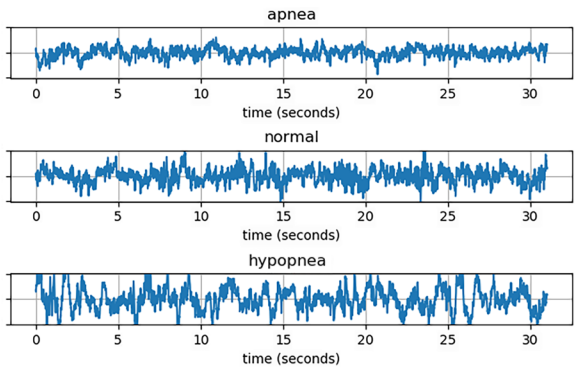


Fig. 8 Three categories of EEG signals

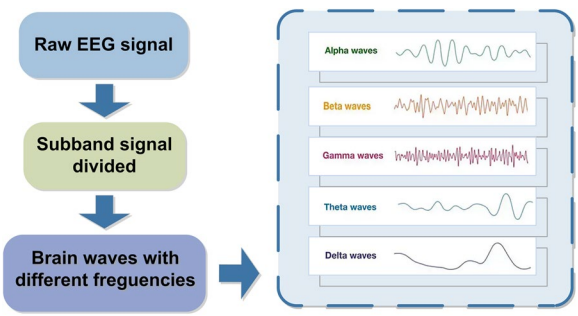


Fig. 9 Pre-processing methods of EEG signal

wave segments and subsequently utilized for automatic diagnosis by an appropriate classifier.

Table 5 summarizes the machine/deep learning techniques for EEG signal processing. It is worth noting that the combination of C3-A2 and C4-A1 channels is commonly utilized. The approaches for single-channel inputs, multi-channel inputs, and multi-modal inputs are separately listed below.

Methods for single-channel inputs

Some studies used single-channel EEG for apnea–hypopnea detection. Mahmud et al. [82] developed a model combining FCNN and DenseNet for sleep apnea diagnosis. The FCNN is utilized for automated feature extraction, while the Dense classifier is employed for apnea identification. The training set consists of 5 patients, and the test set comprises another 5 patients. The Dense classifier finally obtained the best performance with an accuracy of 77.14%. Moreover, a year later, they designed another method, which is a hybrid model based on CNN-BiLSTM [83]. A variational mode decomposition algorithm (VMD) is utilized in the method to decompose the raw EEG signal into multiple modes. FCNN is used to extract features from each VMD mode, and BiLSTM

Table 5 Methods related to EEG

Study	Signals	Events	Methods	Performance(%)	Subjects	Database	Year
[82]	EEG	NH/A	FCNN	Accuracy = 77.14	25	UCDDB	2020
[83]	EEG	N/A	CNN-BiLSTM	UCDDB: Accuracy = 93.22, MIT-BIH: Accuracy = 89.41, PCCC 2018: Accuracy = 93.25	-	UCDDB, MIT-BIH, The PhysioNet Computing in Cardiology Challenge 2018 (PCCC 2018) [84]	2021
[85]	EEG	N/A	EEG-CLNET	Accuracy = 75	60	UCDDB, Local Hospital	2023
[86]	EEG (C3-A2, C4-A1)	N/AH	RF, KNN, SVM	RF: Accuracy = 94.33, KNN: Accuracy = 93.84, SVM: Accuracy = 80.43	30	Tianjing Chest Hospital	2020
[87]	EEG (C3-A2, C4-A1)	NA	BiLSTM	Accuracy = 92.73	30	Tianjing Chest Hospital	2022
[88]	EEG (C3-A2, C4-A1, A1-A2)	AHI Severity, MAD	1-D CNN, GRU	Sensitivity = 73	234	PSG-Audio	2021
[89]	EEG (C3-A2, C4-A1)	OSA, CSA	RF, KNN, SVM	RF: Accuracy = 88.99, KNN: Accuracy = 85.51, SVM: Accuracy = 68.7	30	Tianjing Chest Hospital	2021
[90]	EEG (C4-A1), oral-nasal airflow (thermal, nasal)	N/AH	2-D ResNet	Accuracy = 87.7	50	Private Dataset	2023
[91]	EEG, ECG, airflow	N/A	ResNet, ShuffleNet	UCDDB: Accuracy = 90.45, MIT-BIH: Accuracy = 90.72	41	UCDDB, MIT-BIH	2021

FCNN Fully Connected Neural Network, *MAD* Mean Apnea-Hypopnea Duration, *RF* Random Forest, *KNN* k-Nearest Neighbor, *GRU* Gate Recurrent Unit

is used to optimize the extracted features. The average accuracy across three publicly available datasets was 93.22%.

In [85], an EEG-based collaborative learning network called EEG-CLNet was designed. Local information aggregation network (LIAN) was constructed as the backbone for EEG-CLNet, comprising a bottom feature extraction module (consisting of three 1-D CNN blocks and a feature fusion block), a transformer layer, and a classifier. This model regards different tasks as a common unit to extract features from intragroups via both local parameter sharing and cross-task knowledge distillation (CTKD), rather than just sharing parameters or shortening the distance between different tasks. The approach has been validated on two databases (UCDDB and the local hospital dataset) with equal or better performance than other methods. The accuracy was 74% for the UCDDB and 75% for the local hospital dataset.

Methods for multi-channel inputs

In addition, two channels (C3-A2, C4-A1) of EEG signals are usually used by some works. In [86], the authors proposed a novel approach that combines discrete wavelet transform and approximate entropy of two channels of EEG signals for the detection of sleep apnea–hypopnea events. KNN using two channels of EEG achieved the best performance, with an accuracy of 94.33%. Similarly, in [89], the researchers proposed a novel approach to categorize OSA, CSA and normal breathing events by using two channels of EEG signals. After a 10-fold cross-validation, they found that the average accuracy rate was 88.99%. Specifically, the classification accuracy of OSA, CSA and normal breathing was 80.43%, 84.85%, and 95.24%, respectively. However, in [87], they used BiLSTM model to test the signals of C3-A2 channel and C4-A1 respectively. The results showed that C3-A2 obtained the accuracy of 90.05% and C4-A1 achieved the best accuracy of 92.73%. It can be explained that, during sleep, the right (C4-A1) EEG signal in the central scalp region of patients with sleep apnea changed more significantly than the left (C3-A2) because the right hemisphere is more active than the left hemisphere [109]. Furthermore, authors in [88] proposed a hybrid network called CRNN consisting of three layer 1-D CNN layer and one Bi-directional GRU layer, whose inputs were three channels of EEG signals (C3-A2, C4-A1, A1-A2). In the preprocessing step, three channels of EEG signals were split into 300-second-long, non-overlapping segments. For each segment, the spectrogram was produced and the generated time series data per frequency for each channel were considered as separate input features. Mean apnea–hypopnea duration (MAD) and AHI were estimated in this work with great accuracy, especially in the class of severe apnea.

Methods for multi-modal inputs

Additionally, EEG is always utilized accompanied by other signals (such as nasal airflow and ECG) for apnea detection. In [90], the C4-A1 channel EEG signal and oral-nasal signals were used as inputs of 2-D ResNet

with branch convolution block and self-attention mechanism. The model achieved an average accuracy of 87.7% after conducting a 5-fold cross-validation approach on 50 recordings. Moreover, in [91], this paper proposed a model called RP-CNNs for OSA detection. RP images were created by using EEC, ECG and airflows, and then fed into the models ResNet-18 and shuffleNet. A 10-fold cross-validation approach was used in MIT-BTH and UCDDB. The achieved accuracy of two datasets was 90.72% and 90.45% respectively.

Snoring sound signals

Snoring is caused by the respiratory airflow velocity increases at narrow sections within the upper airways during sleep, which leads to trigging tissue vibrations [110]. It is also the most common symptom of obstructive apnea and hypopnea [111]. Several OSA-related features, such as Mel-frequency cepstral coefficients, 800 Hz power ration, spectral entropy could be obtained from snoring sound data [112]. Due to this reason, numerous automated approaches have been developed for the detection of sleep apnea/hypopnea by utilizing snoring sounds. Additionally, there have been studies [113] focusing on snoring sound features associated with sleep apnea and hypopnea in order to assess the severity of these conditions. The machine/deep learning techniques employed for analyzing snoring sound signals are summarized in Table 6. Methods specifically designed for 1-D inputs, 2-D inputs, and multi-modal inputs are separately listed below.

Methods for 2-D inputs

Some works transformed the raw sleep sound into Mel-spectrogram. In [104], this paper proposed a pre-trained VGG19+LSTM fused model to classify normal and OSAHS patients, and detect apnea–hypopnea snoring event from the whole night data. This model achieved the highest accuracy of 99.31% in two classes classification and 66.29% accuracy in AHI severity detection. In [103], this paper proposed a novel approach to screen for OSA, the whole night snoring sounds were recorded with a in-home smart-phone. A Deep-Learning Neural Network (DNN) was proposed to classify the sound segments, in order to estimate the AHI. The moderate OSA achieved a sensitivity of 79% and a specificity of 80%. The severe OSA achieved a sensitivity of 78% and a specificity of 93%. In [102], this paper proposed a 2-D CNN to identify sleep apnea by using spectrogram of audio signals collected from smartphone. A 60-second-long window obtained the best performance with an accuracy of 88.5%. Similarly, in [101], this paper proposed an automatic OSA detection method based on 2-D CNN called OSAnet by using audio data. They transformed the raw audio data into Mel-spectrogram as the inputs. The model achieved a precision of 81% and a sensitivity of 78% in the test set.

Table 6 Methods related to snoring sound

Study	Signals	Events	Methods	Performance(%)	Subjects	Database	Year
[100]	Snoring sound	N/A	2-D CNN-LSTM	Accuracy = 75.13	105	Shenzhen People's Hospital	2022
[101]	Snoring sound	N/A	2-D CNN, ResNet	Sensitivity = 95.6, Specificity = 91.6	194	Beijing Tongren Hospital	2022
[102]	Snoring sound	N/A	2-D CNN	Accuracy = 88.5	25	Hospital Clinic de Barcelona	2022
[103]	Snoring sound	OSA severity, apnea–hypopnea events detection	DNN	Sensitivity = 78, Specificity = 93	103	HSAT	2022
[104]	Snoring sound	Two classes (normal snoring, abnormal snoring)	VGG19-LSTM	Accuracy = 99.31	50	The First Affiliated Hospital of Guangzhou Medical University, PSG-Audio	2023
[105]	Snoring sound	Three classes (hypopnea, apnea snoring, normal snoring, non-snoring)	FCN	Accuracy=85.25	5	PSG-Audio	2022
[106]	Snoring sound	NS, NSP, PAS	LSTM	Accuracy = 95.3	33	Hangzhou Second People's Hospital	2022
[107]	Snoring sound	Snoring sounds related to apnea events, normal snore	LSTM, CNN	LSTM: Accuracy = 87, CNN: Accuracy = 85	32	Volunteers	2020
[108]	Snoring sound, tracheal motion, SpO_2	N/A, AHI severity	DNN	Accuracy = 92.45	233	University Health Network Sleep Laboratories	2023

NS normal snoring from normal people, *NSP* normal snoring of Obstruvtive Sleep Apnea and Hypopnea Syndrome (OSAHS) patients, *PAS* post-apnea snoring (two second snoring after the end of apnea)

Methods for 1-D inputs

Additionally, some existing works classify snoring sounds more detailly. In [106], a multi-input LSTM was designed to classify three classes of snoring sound data: (1) normal snoring from normal people (NS), (2) normal snoring of OSAHS patients (NSP) and (3) post-apnea snoring, two second snoring after the end of apnea (PAS). The snoring sound data were collected from 33 patients and 10 normal people by Mics. The two classification methods achieved an accuracy rate of 95.3%. Additionally, the three-classification approach yielded an accuracy rate of 81.6%. Similarly, in [105], they also devoted to conduct on three classification including normal snoring, apnea/hypopnea snoring and non-snoring, by utilizing sleep sound data. A deep learning model consisting of three fully connected layers was implemented and got the final accuracy of 85.24%.

In [107], The authors proposed a deep learning model, combining CNN and LSTM, for the classification of snoring sounds associated with apnea events and non-apnea related snoring sounds, as well as predicting the severity of OSAHS. Three feature extraction methods Mel Frequency Cepstral Coefficient (MFCC), LPCC and LPMPCC were used to extract relevant features. The experimental results showed that the combination of MFCC and LSTM achieved the best accuracy of 87%.

Hybrid network is also common in sleep apnea detection. In [100], this paper proposed a SAS classification model based on 2-D CNN-LSTM model. Where, a mel-FBank was used for feature extraction and the snoring sound signal length is 1-second-long with 8820 Hz. After 5-fold cross-validation, the average accuracy is 75.13%.

Methods for multi-modal inputs

The combination of snore data and other signals could be used in sleep apnea/hypopnea detection. In [108], snoring sound data was combined with tracheal motion, SpO_2 for sleep apnea detection. A DNN-based model was designed that consists of two DNNs, one to detect spontaneous (short-term) changes and the other to detect long-term changes that may be presumably associated with sleep apnea events and their iterations. The results showed that $AHI_{PSG} \geq 15$ obtained the highest sleep apnea detection accuracy of 92.45%.

Challenges and future work

In the development of methods for automatic diagnosis of sleep apnea/hypopnea, the challenges need to be addressed and future research directions related to AI techniques are listed below.

Challenges

Limited data availability

Deep learning models require large volumes of high-quality labeled data to effectively train. However, obtaining accurately labeled sleep events in the case of sleep apnea and hypopnea can be challenging due to the time-consuming nature of sleep studies and the need for domain expertise. Moreover, various external factors during data acquisition can impact human physiological signals, thereby limiting data availability. Notably, data collected in a home environment is more susceptible to interference compared to laboratory settings. For instance, background noise in a home environment may weaken snoring sounds from OSA patients [114]. Hence, it is essential to establish specific protocols for collecting data in different environments. Additionally, selecting an appropriate data pre-processing method is crucial.

Imbalanced data problem

In this review paper, we have observed that numerous studies have encountered the challenge of data imbalance. Specifically, sleep apnea and hypopnea events are often infrequent compared to non-event periods during sleep. Moreover, the number of hypopnea events is typically much lower than that of apnea and normal events in a typical dataset. Imbalanced datasets can result in models that exhibit bias toward the majority class and may yield poor performance in detecting minority events. This issue can be addressed from two perspectives. Firstly, data augmentation can be performed by utilizing generative models like GANs and diffusion models. Secondly, employing general algorithms such as adaptive sliding window weighted algorithm [115] and boosting window algorithm [116] can improve the performance of minority classes during training process.

Multi-center study necessity

The present study reveals that the generalizability of findings from single-center studies may be limited due to variations in sleep patterns, physiology, age range, AHI distribution, and patient characteristics [46]. Consequently, these findings might not accurately apply to different demographic groups and could potentially result in inaccurate diagnoses. To overcome this limitation, it is imperative to conduct additional external validations involving diverse populations from multiple centers in both laboratory and home-based settings [101].

Transferability to real-world settings

Models developed in controlled research environments might not perform as well when deployed in real-world clinical settings due to variations in data quality, noise, and patient conditions.

Changing medical guidelines

Medical guidelines and diagnostic criteria can evolve over time. Deep learning models may need constant retraining and adaptation to stay aligned with the latest clinical knowledge.

Future work

The future of AI in sleep apnea/hypopnea detection holds significant promise for enhancing the diagnosis, monitoring, and treatment of this sleep disorder. The integration of AI technology with healthcare providers can pave the way for more accessible, efficient, and personalized care for individuals suffering from sleep apnea/hypopnea.

End-to-end system construction

The future will witness further research on the automatic detection of sleep apnea/hypopnea, necessitating the development of an end-to-end detection system with deep learning as its core. This system should possess data storage capabilities for analysis, anomaly detection and alerting functionalities, as well as the ability to generate comprehensive long-term reports. Such advancements hold significant value in various domains such as geriatric care homes [117].

Multi-task analysis

The utilization of AI for automated analysis of sleep state staging and identification of sleep apnea and other events on PSG and other data enables its application in various medical processes, including assessment, prediction, treatment, and rehabilitation of sleep disorders, depression, and other mental illnesses. Furthermore, it can also provide predictions regarding the likelihood of complications associated with apnea such as cardiovascular problems or cognitive impairment. Early detection of these risks facilitates the guidance of preventive measures and treatment plans.

Telemedicine and Remote monitoring

What is more, within the realm of telemedicine and remote diagnostics, AI techniques can be seamlessly integrated into telemedicine platforms to facilitate remote diagnoses and enable sleep specialists to evaluate cases of sleep apnea/hypopnea from a distance. This is particularly advantageous for patients residing in remote areas or those with limited access to specialized sleep centers.

Conclusion

AI techniques in diagnosing sleep apnea/hypopnea enhance the efficiency, accuracy, and scalability of sleep disorder diagnosis. By combining the expertise of sleep experts with the computational power of deep learning models, healthcare experts can deliver more precise and timely diagnoses, leading to improved patient outcomes and better management of sleep-related disorders. In this article, AI-based automated detection methods, including machine/deep learning methods for sleep apnea/hypopnea using single or multi-modal physiological signals are comprehensively reviewed. We summarize the general process framework of the existing automatic sleep apnea detection methods, including data acquisition, raw signal pre-processing, machine/deep model construction, event classification, and evaluation. And we also summarized commonly used databases and pre-processing methods, mainly including denoising, normalization, image creation and other pre-processing steps. In addition, we summarize and analyze existing works using different physiological signals including nasal airflow, SpO_2, ECG, EEG and snoring sound respectively and draw relevant tables. The content of the form includes signal, events, method, performance, subjects, database and year. Specific signal pre-processing methods based on the characteristics of different physiological signals are also covered. We also describe the challenges faced today, mainly including limited data availability, imbalanced data problem, multi-center study necessity transferability to real-world settings, and changing medical guidelines. And we finally discuss the future research directions such as the end-to-end system construction, multi-task analysis and telemedicine and remote monitoring.

Data availability

Not applicable.

Declarations

Conflict of interest

We declare that we do not have any commercial or associative interest that represents a Conflict of interest in connection with the work submitted.

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